Model-Theoretic Quantifier Elimination

A Lean formalization over `mathlib4`

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Model Theory

What is Model Theory?

- Model theory is a branch of mathematical logic where we study mathematical structures by considering the first-order sentences true in those structures and the sets definable by first-order formulas.
- Given concrete mathematical structures, we can obtain new information about the structure using model-theoretic techniques.

Language

- In mathematical logic, we use first-order languages to describe mathematical structures.
- A $language$ $\mathcal{L}$ consists of:
  - a set of function symbols $\mathcal{F}$
  - a set of relation symbols $\mathcal{R}$
  - a set of constant symbols $\mathcal{C}$
- Examples:
  - The language of rings $\mathcal{L}_r = \{+, -, \cdot, 0, 1\}$
  - The language of orders $\mathcal{L}_o = \{<\}$

Structure

- A structure interprets the symbols of a language on a domain.
- Formally, an *$\mathcal{L}$-structure* $\mathcal{M}$ consists of:
  - a non-empty set M (the domain/universe),
  - for each function symbol $f \in \mathcal{F}$, a function $f^{\mathcal{M}} : \mathcal{M}^n \to \mathcal{M}$,
  - for each relation symbol $R \in \mathcal{R}$, a relation $R^{\mathcal{M}} \subseteq \mathcal{M}^n$,
  - for each constant symbol $c \in \mathcal{C}$, an element $c^{\mathcal{M}} \in \mathcal{M}$.
- Examples: $(\mathbb{N}, 0, 1, +, \cdot)$, $(\mathbb{Q}, \leq)$.

Terms

- An *$\mathcal{L}$-term* is:
  - a constant symbol c,
  - a variable v_i
  - or a function symbol f applied to terms $t_1, \ldots, t_n$.

Formula and Sentence

- An *atomic $\mathcal{L}$-formula* is a relation symbol applied to terms, or an equality between terms.
- *$\mathcal{L}$-formulas* are built from atomic formulas using logical connectives and quantifiers.
- An *$\mathcal{L}$-sentence* is a formula with no free variables.
- Example: $\exists z\,(x < z \wedge z < y)$ is formula but not a sentence, as it has free variables x and y.

Satisfaction

- We say that an $\mathcal{M}$ *satisfies* a formula φ, written $\mathcal{M} \models \varphi$, if φ holds in $\mathcal{M}$.
- Formally, the satisfaction relation is defined inductively on the structure of formulas.
- Example: $(\mathbb{Q}, <) \models \forall x \forall y \, (x < y \implies \exists z \, x < z \wedge z < y)$.

Theory and Model

- An *$\mathcal{L}$-theory* is a set of $\mathcal{L}$-sentences.
- We say that a structure $\mathcal{M}$ is a *model* of a theory T, written $\mathcal{M} \models T$, if $\mathcal{M}$ satisfies every sentence in T.
- A theory is *satisfiable* if it has a model.
- Moreover, we say that a theory T satisfies a sentence φ, written $T \models \varphi$, if every model of T satisfies φ.

Quantifier Elimination

- A theory T has *quantifier elimination* if for every formula φ, there is a quantifier-free formula ψ such that

$$T \models \varphi \leftrightarrow \psi$$

Dense Linear Orders without Endpoints

- Let $\mathcal{L}_o = \{<\}$ be the language of orders.
- Let DLO be the $\mathcal{L}_o$-theory consisting of the following sentences:

$\forall x \neg(x < x)$	(Irreflexivity)
$\forall x \forall y \forall z \, (x < y \wedge y < z \rightarrow x < z)$	(Transitivity)
$\forall x \forall y \, (x < y \vee x = y \vee y < x)$	(Linearity)
$\forall x \forall y \, (x < y \rightarrow \exists z \, (x < z \wedge z < y))$	(Density)
$\forall x \exists y \exists z (y < x \wedge x < z)$	(No endpoints)

- Examples of models of DLO: $(\mathbb{Q}, <)$, $(\mathbb{R}, <)$, $((0,1), <)$ with the usual order.

Does DLO have Quantifier Elimination?

- DLO has quantifier elimination.
- Every formula in the language $\{<\}$ is equivalent over DLO to a quantifier-free formula.
- Examples:

$$\begin{aligned} \exists z\,(x < z \wedge z < y) &\iff x < y \\ \exists z\,(x < z \wedge w < z \wedge z < y) &\iff x < y \wedge w < y \\ \exists z\,\big(x < z \wedge z < y \wedge \neg(z < w)\big) &\iff x < y \wedge (w < y \vee w = y) \end{aligned}$$

Type Theory

Type Theory

- We work in the sense of Martin-Löf Type Theory: a constructive foundation of mathematics.
- The basic syntactic form is a *judgement*, not a proposition.
- Every object and expression has a (unique) type, statically determined.
- This is the foundation used by proof assistants such as Lean, Agda, and Coq.

proofs as programs propositions as types

Type Theory vs Set Theory

Set theory

- $3 \in \mathbb{N}$ is a *proposition*.
- Membership is the basic relation.
- Functions are derived: subsets of $A \times B$ with certain properties.
- One may ask encoding-dependent questions, e.g. $\mathbb{N} \cap \mathrm{Bool} = \emptyset$?

Type theory

- $3 : \mathbb{N}$ is a *judgement* (static information).
- Typing is the basic judgement; $a : A$ is not a proposition.
- Functions are primitive: given $A, B : \mathrm{Type}$, we form $A \to B : \mathrm{Type}$.
- Type theory is extensional in the sense that we cannot talk about implementation details.

Judgements

- A *judgement* is static information that the type system can check.
- The central judgement is $a : A$: "a is an element of type A".
- A *context* Γ records the assumptions currently in scope.

$$\Gamma = x : A,\ f : A \to B \ \vdash\ f\,x : B$$

Functions

- Given $A, B : \mathrm{Type}$, the function type $A \to B : \mathrm{Type}$ is primitive.
- A function is defined by its action on inputs, e.g. $f : \mathbb{N} \to \mathbb{N}$ with $f\,x \equiv x + 3$.
- Anonymous functions use λ-abstraction: $\lambda x.\, x + 3 : \mathbb{N} \to \mathbb{N}$.

$$(\lambda x.\, x + 3)\, 2 \equiv 2 + 3 \qquad (\beta\text{-reduction})$$

- Every function has exactly one argument; multiple arguments use currying: $g : \mathbb{N} \to (\mathbb{N} \to \mathbb{N})$, so $g\,3\,2 : \mathbb{N}$.
- Application associates to the left; $\to$ associates to the right.

Products and Sums

- Given $A, B : \mathrm{Type}$, we form the product $A \times B$ and the sum $A + B$.
- $(a, b) : A \times B$ if $a : A$ and $b : B$.
- $\mathrm{left}\, a : A + B$ if $a : A$; $\mathrm{right}\, b : A + B$ if $b : B$.
- The unit type 1 has a single element `()`; the empty type 0 has no elements.

Propositions as Types

- A proposition P is represented as a type $P : \mathrm{Type}$.
- A proof of P is a term $p : P$, evidence for the proposition.

to prove P is to construct $p : P$

Curry–Howard Translation

$$
\begin{aligned}
[\![P \Rightarrow Q]\!] &\equiv [\![P]\!] \to [\![Q]\!] \\
[\![P \wedge Q]\!] &\equiv [\![P]\!] \times [\![Q]\!] \\
[\![\mathrm{True}]\!] &\equiv 1 \\
[\![P \vee Q]\!] &\equiv [\![P]\!] + [\![Q]\!] \\
[\![\mathrm{False}]\!] &\equiv 0 \\
[\![\forall x : A.\, P(x)]\!] &\equiv \Pi x : A.\, [\![P(x)]\!] \\
[\![\exists x : A.\, P(x)]\!] &\equiv \Sigma x : A.\, [\![P(x)]\!]
\end{aligned}
$$

Dependent Types

- A *dependent type* is a family $B : A \to \mathrm{Type}$ indexed by A.
- Π-types generalise function types: the codomain may depend on the domain.
- Σ-types generalise products: the second component may depend on the first.

$$\Pi x : A.\, B(x) \qquad \Sigma x : A.\, B(x)$$

- Example: $\mathrm{zeroes} : \Pi n : \mathbb{N}.\, A^n$ – assigns to each length n a tuple of n elements of type A.
- Example: $\mathrm{List}\, A \equiv \Sigma n : \mathbb{N}.\, A^n$.
- Predicates are dependent types: e.g. $\mathrm{Prime} : \mathbb{N} \to \mathrm{Type}$.

Typed Predicate Logic

- We use a *typed* predicate logic: quantifiers range over a type A.
- Other connectives are defined: $\neg P \equiv P \to 0$,
  $P \Leftrightarrow Q \equiv (P \to Q) \times (Q \to P)$.
- Not all classical tautologies hold, e.g. $\neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$.
- The law of excluded middle $P \vee \neg P$ is not provable in general.

$$\exists x : A.\, P(x) \;\rightsquigarrow\; \Sigma x : A.\, P(x) \;=\; \text{a pair } (a, p) \text{ with } a : A,\; p : P(a).$$

- A proof of $\exists x : A.\, P(x)$ provides an explicit witness $a : A$ and evidence $p : P(a)$!

Example: Distributivity of Conjunction over Disjunction

Goal: $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$.

For $P, Q, R : \mathrm{Type}$, define maps

$$f : P \times (Q + R) \to (P \times Q) + (P \times R),$$
$$g : (P \times Q) + (P \times R) \to P \times (Q + R)$$

by pattern matching on constructors:

$$f\,(p, \mathrm{left}\,q) \equiv \mathrm{left}\,(p, q), \qquad g\,(\mathrm{left}\,(p, q)) \equiv (p, \mathrm{left}\,q),$$
$$f\,(p, \mathrm{right}\,r) \equiv \mathrm{right}\,(p, r) \qquad g\,(\mathrm{right}\,(p, r)) \equiv (p, \mathrm{right}\,r)$$

The pair (f, g) is evidence for the logical equivalence!

The Equality Type

- Given $a, b : A$, the equality type $a =_A b : \mathrm{Type}$ is generated by one constructor:

$$\mathrm{refl} : \Pi x : A.\, x =_A x$$

- A proof $p : a =_A b$ is evidence that a and b are equal.
- From refl we derive symmetry, transitivity, and congruence:

$$\mathrm{sym} : \Pi x, y : A.\, x =_A y \to y =_A x$$

$$\mathrm{trans} : \Pi x, y, z : A.\, x =_A y \to y =_A z \to x =_A z$$

Dimensions of Types

- The type A with its equality types forms a *groupoid*: refl is identity, sym gives inverses, trans composes paths.
- Types are classified by *dimension* (truncation level): how much higher equality structure they carry.

Dimension	Name
-2	contractible types
-1	propositions
0	sets
1	groupoids

- A *proposition* has at most one element; a *set* has propositional equalities; a *groupoid* has set-like equalities between paths.

Extensionality and Univalence

- Example: $1 + 2$ and $2 + 1$ are indistinguishable, yet not definitionally equal.

A map $f : A \to B$ is an *equivalence* if it has a two-sided inverse up to equality. Write

$$A \simeq B :\equiv \Sigma f : A \to B.\, \mathrm{isEquiv}\, f.$$

Path induction gives $\mathrm{eq2equiv} : A =_{\mathrm{Type}} B \to A \simeq B$.

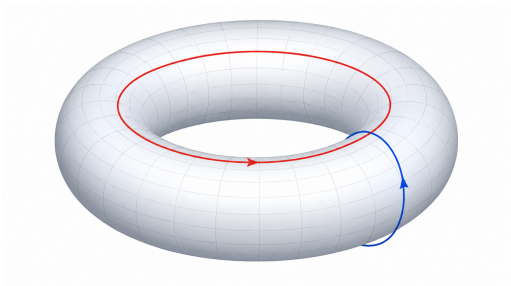
Univalence axiom: $\mathrm{eq2equiv}$ is itself an equivalence:

$$\mathrm{isEquiv}(\mathrm{eq2equiv}) \qquad \iff \qquad (A \simeq B) \simeq (A =_{\mathrm{Type}} B)$$

Consequence: $A \simeq B \to A =_{\mathrm{Type}} B$ – equivalent types may be treated as equal.

Towards Homotopy Type Theory

- Types carry higher-dimensional structure: paths, paths between paths, …
- This ω-groupoid view connects logic, computation, and topology.
- Lean's core is based on intensional type theory; univalence and HoTT are active research areas.



Formalization in Lean

Lean4

- Interactive theorem prover / functional programming language
- Dependent type theory
- Propositions as types, proofs as terms
- UEP / K eliminator
- Mathlib4

Dense Linear Orders

```
/-- The theory of dense linear orders without endpoints. -/
def dlo : L.Theory :=
  L.linearOrderTheory ∪
    {L.noTopOrderSentence, L.noBotOrderSentence,
      L.denselyOrderedSentence}
```

Quantifier Elimination

```
/-- A formula is equivalent to a quantifier-free formula over a theory. -/
def IsQFEquivalent (T : L.Theory) {α : Type*} (φ : L.Formula α) : Prop :=
  ∃ ψ : L.Formula α, ψ.IsQF ∧ φ ⇔[T] ψ

/-- A theory has quantifier elimination if every formula is equivalent,
over the theory, to a quantifier-free formula. -/
def HasQuantifierElimination (T : L.Theory) : Prop :=
  ∀ {α : Type} (φ : L.Formula α), T.IsQFEquivalent φ
```

Marker 3.1.4

Theorem 3.1.4 *Suppose that $\mathcal{L}$ contains a constant symbol c, T is an $\mathcal{L}$-theory, and $\phi(\bar{v})$ is an $\mathcal{L}$-formula. The following are equivalent:*

i) There is a quantifier-free $\mathcal{L}$-formula $\psi(\bar{v})$ such that $T \models \forall \bar{v}\,(\phi(\bar{v}) \leftrightarrow \psi(\bar{v}))$.

ii) If $\mathcal{M}$ and $\mathcal{N}$ are models of T, $\mathcal{A}$ is an $\mathcal{L}$-structure, $\mathcal{A} \subseteq \mathcal{M}$, and $\mathcal{A} \subseteq \mathcal{N}$, then $\mathcal{M} \models \phi(\bar{a})$ if and only if $\mathcal{N} \models \phi(\bar{a})$ for all $\bar{a} \in A$.

```
/-- A formula is equivalent over `T` to a quantifier-free formula iff its truth is invariant under
pairs of embeddings from a common structure into nonempty models of `T`.

This corresponds to [Theorem 3.1.4][marker2002]. -/
theorem isQFEquivalent_iff_realize_iff_of_embeddings
    {T : L.Theory} {α : Type u'} (φ : L.Formula α) :
    T.IsQFEquivalent φ ↔
      (∀ {M N A : Type (max u v u')} [L.Structure M] [L.Structure N] [L.Structure A]
        [T.Model M] [T.Model N] [Nonempty M] [Nonempty N]
        (f : A ↪[L] M) (g : A ↪[L] N)
        (a : α → A), φ.Realize (f ∘ a) ↔ φ.Realize (g ∘ a)) := ...
```

Henson 7.11

7.11. Theorem. *Let Σ be a set of L-sentences. The following conditions are equivalent:*

(1) *Σ has QE.*

(2) *Whenever $\mathcal{M}, \mathcal{N}$ are models of Σ, f is in $\mathrm{Sub}(\mathcal{M}, \mathcal{N})$, and $a \in M$, there exists an elementary extension $\mathcal{N}'$ of $\mathcal{N}$ and a function g in $\mathrm{Sub}(\mathcal{M}, \mathcal{N}')$ such that g extends f and $a \in \mathrm{dom}(g)$.*

(3) *Whenever $\mathcal{M}, \mathcal{N}$ are ω-saturated models of Σ, either $\mathrm{Sub}_0(\mathcal{M}, \mathcal{N})$ is empty or it is a back-and-forth system from $\mathcal{M}$ to $\mathcal{N}$.*

```
/-- If every pair of nonempty models of `T` has the elementary extension-pair property, then `T`
has quantifier elimination.

The hypothesis is condition (2) in [Theorem 7.11][vandendries_henson_2016]; this theorem proves
the implication from that extension property to condition (1). -/
theorem hasQuantifierElimination_of_isElementaryExtensionPair
    {T : L.Theory}
    (h : ∀ {M N : Type (max u v)} [L.Structure M] [L.Structure N]
      [T.Model M] [T.Model N] [Nonempty M] [Nonempty N],
      L.IsElementaryExtensionPair M N) :
    T.HasQuantifierElimination := ...
```

DLO Has Quantifier Elimination

```
/-- The theory of dense linear orders without endpoints
has quantifier elimination. -/
theorem dlo_hasQuantifierElimination :
    Language.order.dlo.HasQuantifierElimination := by
  apply hasQuantifierElimination_of_isElementaryExtensionPairFG
  intro M N _ iN _ _ _ _ f a
  obtain ⟨g, ha, hfg⟩ := Language.dlo_isExtensionPair M N f a
  refine ⟨N, iN, ElementaryEmbedding.refl Language.order N, g, ha, ?_⟩
  -- Mapping the codomain along the identity embedding is definitionally trivial.
  exact hfg.imp fun _ h => h
```

Thank you for listening!

Scan the QR code to checkout our PR in `mathlib4` GitHub repository!